King Nala's Grand Feast

The Puzzle of the Thousand Flames

The Story

Long ago in ancient India...

In the kingdom of **Nishadha**, there was a special kitchen called **The Thousand Flames**. Normal cooks use one fire. But this kitchen had *thousands of small flames working together*—each flame did one small job, but together they made food so good that even the gods loved it.

King Nala was the master of this kitchen. The fire god had given him a secret number called the **Spice Key** K. When Nala used this number in his cooking formula, even the simplest food became amazing.

Tomorrow is the **Royal Wedding Feast** for Princess Damayanti. There will be $N =$ **4,194,304** guests (that is 2^{22}). Each guest must be served a personally crafted dish. Nala's formula gives each guest a **satisfaction score** using whole-number arithmetic:

For guest number i (where i goes from 0 to $N-1$):

$$\text{raw}[i] = (i \times 37 + 13) \bmod 101$$

$$\text{cooked}[i] = (\text{raw}[i] \times K + 59) \bmod 1009$$

$$\text{satisfaction}[i] = \text{cooked}[i] \bmod 101$$

But the evil spirit **Kali** entered Nala's dreams and **erased his memory of** K. The only clue is a **copper plate** that records the **total satisfaction of all guests** from the last feast. Each of Nala's students received a different plate, linked to a different value of K.

You are one of Nala's students. Nala has sent you a message:

"The secret K is a whole number between 1 and 10,00,000 (10 Lakh). Test each one against all 2^{22} guests. Find which K produces the total on YOUR copper plate. Save the feast!"

Your Identity

Your Student ID: _____ **Last 4 digits:** _____

If your last 4 digits have leading zeros, treat them as a number. Example: ID ending in `0042` means last 4 digits = 42.

Your Secret Key (this is what you must rediscover):

$$K_{\text{true}} = 1 + \Big((\text{last 4 digits}) \times 101\Big) \bmod 1000000 = _____$$

Your Target Total T: Run the formula with your K_{true} for all 2^{22} guests, sum all satisfaction scores. That sum is your T.

Example: ID = `20251602012` $\Rightarrow$ last 4 digits = 2012

- $K_{\text{true}} = 1 + (2012 \times 101) \bmod 1000000 = 1 + 203212 = 203213$
- $G = (2012 \bmod 1000) + 1 = 12 + 1 = 13$

Kernel Requirements

You must create the following kernels:

Kernel 1: `computeSatisfactionSum`
- **Purpose:** Compute satisfaction for all 2^{22} guests and produce partial sums
- **Input:** The candidate key K_c
- **Output:** An array of partial sums (one per block)

Kernel 2: `countUnhappyGuests`
- **Purpose:** Count guests with satisfaction below 40
- **Input:** The discovered key K
- **Output:** An array of partial counts (one per block)

Finding the Lost Key

Your Task

Write a program that searches for K.
For each candidate K_c from 1 to 10,00,000, compute the satisfaction of all 2^{22} guests, sum them, and check if the sum equals your target total T. Report the K_c that matches.

In your answer, show:

a. Your Student ID, your K_{true}, and your target total T.

b. The value of K your program found, plus the count of candidates with totals greater than T and less than T.

c. Satisfaction for **Guest** $\#G$ using your K:

Where, your guest number (G):

$$G = \Big(\text{last 4 digits mod } 1000\Big) + 1 = \underline{\hspace{3cm}}$$

- $\text{raw}[G] = (G \times 37 + 13) \bmod 101 = \underline{\hspace{3cm}}$
- $\text{cooked}[G] = (\text{raw}[G] \times K + 59) \bmod 1009 = \underline{\hspace{3cm}}$
- $\text{satisfaction}[G] = \text{cooked}[G] \bmod 101 = \underline{\hspace{3cm}}$

d. The **execution time** of your program.